

Conversations Sentiment and Intent Categorization Using Context RNN for Emotion Recognition

Rifqi Majid

Faculty of Computer Science
Dian Nuswantoro University
Semarang, Indonesia
rifqi.majid29@gmail.com

Heru Agus Santoso

Faculty of Computer Science
Dian Nuswantoro University
Semarang, Indonesia
heru.agus.santoso@dsn.dinus.ac.id

Abstract—Nowadays, a chatbot is often found in daily life, and its role in business becomes more important. We develop Dinus Intelligent Assistant (DINA) chatbot for assisting student administration services. While having a text-based conversation, recognizing emotions in textual-based content is a challenging task. This study uses dataset which contains conversation dialogues. In this study, we preprocess conversation using sentiment analysis then collect the data by its conversations and sentiment analysis results. Then we apply RNN to categorize the emotions based on current conversations. The result of this approach is promising, with a precision accuracy of 0.76. From the result, this algorithm can indeed help recognize emotions from text-based conversations.

Keywords— *Deep Learning, RNN, Chatbot, Sentiment Analysis, NLP*

I. INTRODUCTION

During the Industrial Revolution 4.0, Artificial intelligence (AI) was one of the driving pillars of automation and data transfer via the internet [1]. The role of AI is increasingly significant, particularly in the modeling, controlling, and optimizing business services or in the industrial field of manufacture [2]. As one branch of AI, machine learning (ML) is a method that is widely used lately. Machine Learning can be applied in various ways. One of them is applying machine learning in Natural Language Processing (NLP) research field. In the development of the robot, especially chatbot, NLP is needed to understand unstructured textual data's content and context to recognize emotions in contextual chatbot conversations.

In developing intelligent robots or chatbots with AI as a pillar of the industrial revolution, there is a problem identifying emotions contextually. Identification is needed to find out the service satisfaction an organization provides when users interact with smart chatbots. For example, a student interacts with a chatbot to get some information he needs. The conversations that occur can be quite long or short, depending on each situation. But in the conversation, some feelings or emotions are contained during students' interactions with a chatbot [3].

Student

Halo min apakah ada alternatif lain saat upload bukti pembayaran via fax? Karna sulit nemuin fax disini. *(disgust)*

University Staff

Halo. Untuk upload bukti pembayaran pendaftaran online dapat dilakukan langsung melalui sistem, jadi tidak perlu menggunakan fax. Terima kasih. *(neutral)*

Figure 1. Example of communication via guestbook

Shown in Fig 1 is an example of a conversation from Dian Nuswantoro University guest book. It demonstrates contextual emotions containing in a textual-based conversation. There are two kinds of emotional expression in the conversation. The interactions of a latent variable may lead to diverse sentiment dynamics in the conversations [4].

Many studies have been conducted to overcome identifying emotions in chatbot conversations with deep learning or machine learning. Identification of feelings or emotions can be carried out using the Neural Network algorithm in context-aware text.

One of emotional extraction research is done by studying the textual-document contextually, through analyzing the emotions. The emotions are then represented in their textual conversation to users. This method is done by analyzing semantic texts from tweets or chats that do not have a label. It also identifies the relationship of textual information using sentiment analysis techniques [5].

Deep learning methods have gained prominence recently. Many studies have been conducted to identify feelings, emotions, and personalities during conversations. The study to extract the personality of people using the Convolutional Neural Network (CNN) algorithm from an essay written has been done. In this study, the authors combine the vectors obtained from text extraction directly in the preprocessing stage [6].

Then there is also a study that proposes a technique of so-called sent2affect as an adaptive and effective form of computational learning. The performance evaluation is evaluated in six benchmark datasets where the results are obtained, that the recurring neural network and transfer learning consistently perform better than machine learning [7].

DINA chatbot is a robot developed for higher education academic services, including educational services, to ensure prospective students' satisfaction [3]. This research proposes the development of a chatbot using a structured knowledge-base extracted from textual documents. The chatbot acts as a conversational agent that can serve as a virtual administration staff to prospective candidates. A conversation about student administration services is analyzed using ontology as background knowledge combined with learning techniques using the multi-layer perceptron (MLP). However, this kind of traditional neural network has many obstacles in our previous study when identifying emotions from textual conversations via chatbot. Our previous approach also has low accuracy when dealing with emotions or feelings identification.

Based on the literature study results above, emotional extraction of conversation on chatbots has been conducted using various methods. In this study, we first preprocess the dataset using sentiment analysis, collect the sentiment result, and provide benefits to analyze conversations. Based on our literature study, RNN is suitable for processing a sequential conversation. RNN is used to train the datasets and then transfer its parameters to the target model. This model fits DINA Chatbot, which needs to identify or recognize user emotions from the textual-based conversation.

II. METHOD

A. Sentiment Analysis

Generally, the sentiment analysis method is the study of extracting people's sentiment of subjective text systematically [12]. A set of features extracted from textual documents uniquely describes the properties possessed by the data. Machine learning algorithms use a number of dimensional features to extract knowledge, including the sentiment. In case the data has high dimensional features, it can be mapped into lower-dimensional features. It must preserve the information that resides in the data. Generally, sentiment analysis employs three main Natural Language Processing (NLP) approaches. They are:

- 1) Rule-based: an approach that based on a set of manually created rules.
- 2) Automatic: an approach that uses artificial intelligence or machine learning to learn patterns from training set.
- 3) Hybrid: an approach that combine automatic and rule-based approach.

Sentiment analysis is the method to capture human feelings, senses, and emotions extracted from textual data. The technique is extensively used in social media analytics, business analytics, data mining, etc. As a method to judge human behavior, sentiment analysis creates opportunities for societies, businesses, and research. Human sentiment can be neutral, negative, or positive represented by numerical scores articulating how much weight of their sentiment. The calculation of sentiment also produces human character toward a particular topic and situation. Hence, sentiment analysis and mining of opinion are often used interchangeably in many cases. However, they also express some dissimilarity among themselves. Sentiment analysis and opinion mining try to discover opinions, grouping the attitude they convey and

categorized them into three possible classes mentioned above, that are positive, negative, and neutral

Generally, the different sentiment classification approach consists of two main methods, i.e., the machine learning and the lexicon-based approach. The machine learning approach can be further categorized into three learning methods, supervised, unsupervised, and reinforcement learning. Supervised learning is the process of learning from the dataset contained target attributes, i.e., positive, negative, and neutral, to generate a model initially. It then further predicts the target class for unseen data. On the other hand, unsupervised learning categorizes unlabeled datasets to predict class of the provided input data. The method of sentiment analysis is depicted in Figure 2.

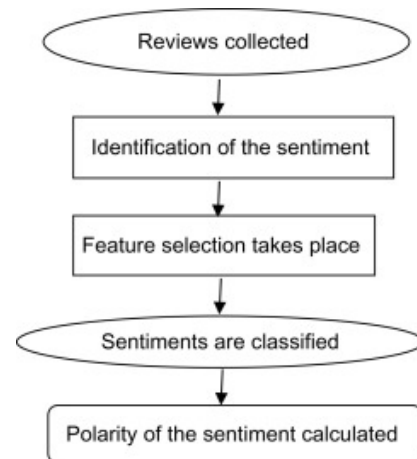


Figure 2. Sentiment Analysis Process

B. Recurrent Neural Network (RNN)

As one of the deep learning algorithms, RNN extends the architecture of traditional NN by increasing the number of hidden layer components. The difference is that RNN introduces the concept of memory, keeping previous information available in the memory. It exists for further processing stages. Hence, it takes into account the sequence of data. Unlike backpropagation of NN, the outputs of hidden layers are also fed back again into the current hidden layer. The addition of previous information allows the analysis of sequential data. This algorithm is a strategy that NN incapable.

Furthermore, NN is limited to the particular number of input, whereas the RNN does not have that restriction. For time-series data containing repeated patterns, RNN recognized them and took into account the benefit of time-related contextual information. This is fulfilled by applying more weight to forms where the previous and following data chunks are identified instead of being evaluated in isolation. For example, a system that learns to identify spoken languages. This kind of system would benefit very much by adding recently spoken word information for next sentence prediction. The inclusion of links between layers in the reverse direction provides feedback loops, to assist learning concepts based on context. This can do for a deep learning application. This phase depends highly on the requirements (or goal) and the data.

The main purpose of the recurrent neural network approach is to provide a fixed-length vector of the text variable [8]. These models generally consist of several layers that have

different functions. They are (1) the projection layer, which is used to maps words, (2) sub-word units or n-grams that are used as vector representations (often trained previously using an unsupervised approach), and then (3) the function of combining them with the different network architectures. It will produce a new appropriate model for the problem.

There are different type of models to represent a textual document conversation, such as Neural Bag-of-Words (NBOW) and transfer learning using RNN [9]. On the other hand, recursive neural network as also known as RecNN, and convolutional neural network (CNN) is also used in many studies. Precisely, these methods take input a set of words extracted from the text sequences, and then represented their meaning with a fixed-length vector. As reported in these studies, RNN performs better performances when it is carried out over a textual dataset. Therefore, RNN scheme is considered as the most suitable architecture for NLP problems. Furthermore, these studies also concluded that the recurrent structure of RNN is very appropriate to process high-dimensional textual datasets.

RNNs come with different types of strategies depending on the particular task they are given. Different type of RNN is distinguished by the number of input components, and also the different number of outputs. The four different types of RNN are:

- 1) One-to-one, also known as the Vanilla RNN. This type of RNN has an input. It can be a word or an image, whereas the output is a single token, such as a word or a boolean value.
- 2) One-to-many, a type of RNN where an input is employed to produce several outputs.
- 3) Many-to-one, a type of RNN where several inputs are utilized to produce a single output.
- 4) Many-to-many, a type of RNN where several inputs generate several outputs.

C. Sentiment Analysis

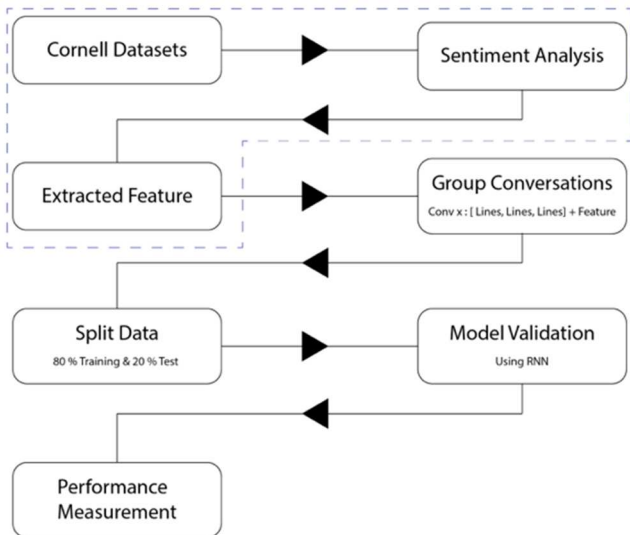


Figure 2. Blockchart process of emotion extraction

This study uses RNN to categorize the intent of textual conversation to obtain user emotions. Therefore, type of RNN we use is many to one.

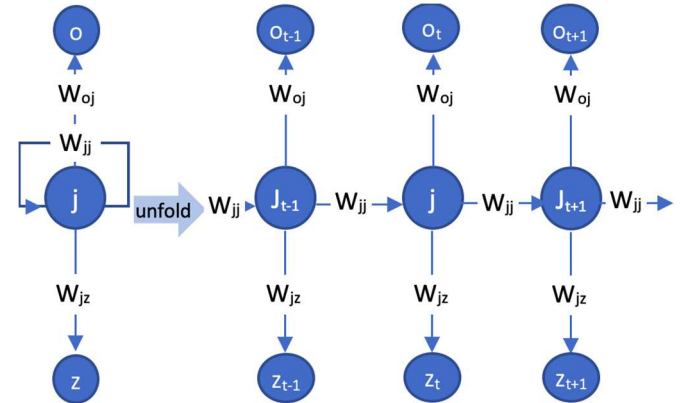
Our proposed approach is depicted in Figure 2. The approach consists of two main modules that are (1) conversation sentiment analysis to extract the features and (2) emotion classification using RNN based on conversation and its sentiment. In this study, we extracted three main features, Positive, Negative, and Neutral. This feature is then attached to each conversation through conversation id.

RNN consists of a few layers with repetitions. This model is suitable for sequence data, data in the form of rows, historical data based on periods, text in the form of word sequences, letters, sub-words, sentences, sounds in the form of frequency, amplitude, and video, which consists of a series of images which are commonly called sequences of image frames. With these few layers, RNN is categorized as deep learning because of the loop in the process [6].

Furthermore, RNNs are appropriately modeled to handle a series of sequential data type, including time-series data. RNN enables data to flow from forward, followed by backward directions. It enables loops in an epoch, and provide links between hidden components. With the aid of such a link scheme, RNN is an appropriate method for utilizing any information from the past data to perform prediction. In other words, RNN provides the facility to capture relationships between data to each other. On the other hand, the backpropagation neural networks only enables data or information to flow forward from input components to output components.

Figure 3. RNN architecture

Figure 3 depicts the scheme of RNN. Given z as a time series



data input, where $z = \{z_1, z_2, \dots, z_T\}$, RNN algorithm computes the sequence of hidden state j , where $j = \{j_1, j_2, \dots, j_T\}$. The output sequence $o = \{o_1, o_2, \dots, o_T\}$ is then iteratively computes using the following set of equations:

$$J_t = f(W_{jz}z_t + W_{jj}j_{t-1} + b_j) \quad (1)$$

$$y_t = g(W_{oj}j_t + b_o) \quad (2)$$

In equations (1) and (2), W_{jz} , W_{jj} , and W_{oj} denote the weight of hidden-input, hidden-hidden weight, and also hidden output weight matrix, respectively. The vectors b_h and b_y representing the bias of hidden layer and output layer, respectively. RNN employs the hidden state h_t at the time step t to memorize the network. In this scenario, the hidden state captures and stores any information included the information in the prior time steps. The function $f(\cdot)$ and $g(\cdot)$ are the activation functions use in the hidden layer and output layer. For time series forecasting

application, RNN expresses multistep dependencies as it predicts out-of-sample z_{t+j} data, which rely on data observed at earlier times t_e , where $t_e < t+j$. However, when the gap of data extends, RNNs lead to suffering increasingly from gradient vanishing problems. In other words, the effect of data at t_e to the predicted data z_{t+j} will decay quickly at time $t+j-t_e$. Therefore, RNN in the simple form may not produce the best option for prediction problems regarding long-term dependencies.

Furthermore, the data were validated by the k fold cross-validation technique. K-Fold Cross-Validation (kFCV) is a model validation technique. The data set D is randomly divided into k sub-stores, which are usually called folds. Data divided can be used (k - 2) fold into training data, one fold for validation, and the other for test data. For example, for k = 5, the D1 data set consisting of three folds contains f1, f2, and f3, which are training data, and for f4, it is validation data while f5 is the test data. The data set D2 consists of f1, f2, and f4 for training data, f3 for validation data, and f5 for test data. D3 data consists of f1, f3, and f4 for training data, f2 for validation data, and f5 for test data. D5 data consists of f2, f3, and f4 for training data, f1 for validation data, and f5 for test data. F5 is deliberately used as test data so that four deep learning models are obtained with one fixed test data. This study uses k = 5. In our proposed approach, each of the turns in the conversation, the results generated from the context RNN, is then projected to the label-space. This step aims to provide emotion prediction to the associated expressions. Our approach trains all sentence expressions in the conversation.

III. EXPERIMENT RESULT

As explained above, this study uses a Cornell movies dataset. The dataset contains conversations inside a movie. This dataset contains 9990 lines of conversations. In this study, we preprocess conversation using sentiment analysis then collect the data by its conversations and sentiment analysis results. Then we apply RNN to categorize the emotions based on current conversations. The result of this approach is promising, with a precision accuracy of 0.76.

Training and Validation Loss

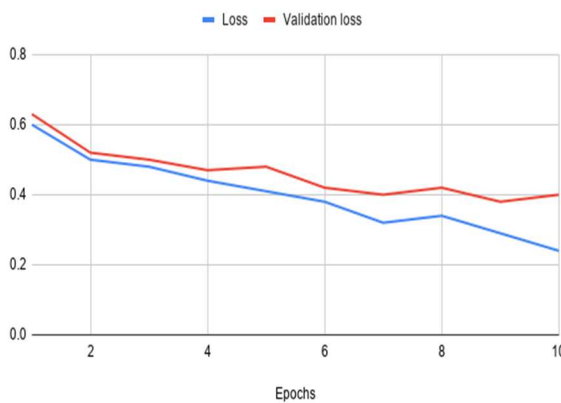


Figure 4. Training and Validation Loss

Training and Validation Accuracy

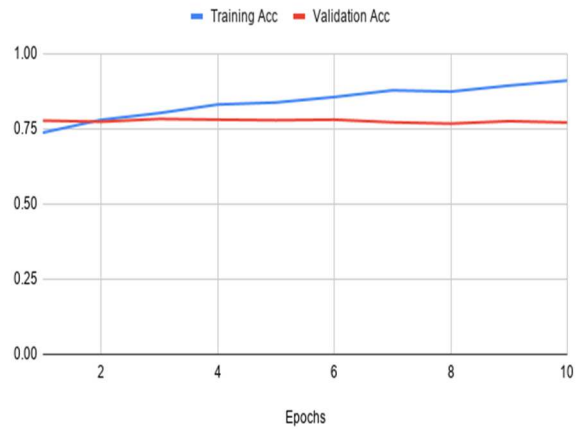


Figure 5. Training and Validation Accuracy

IV. CONCLUSION

Figure 4 and Figure 5 depict the training and validation accuracy and loss. Based on the results obtained above, it can be concluded that emotion recognition using the Context RNN classifications model can obtain the highest precision accuracy of 0.76 with an average loss of 0.4 with the number of epochs is 10. This model will make further DINA chatbot development become more comfortable to recognize user satisfaction based on conversations.

V. SUGGESTION

Future research is expected to improve and stability of the model, so it becomes a more reliable improvement. There remains scope for improvement to achieve more stable training.

Acknowledgment

The author would like to gratefully acknowledge the Director General of Higher Education of Republic Indonesia under the Research Grand with contract number: 018/LL6/PG/SP2H/PL.II/2020

References

- [1] A. Ustundag and E. Cevikcan, Industry 4.0., 2018.
- [2] Azizi and Aydin, Application of Artificial Intelligence Techniques in Industry 4.0, Singapore: Springer Nature Singapore Pte Ltd., 2019.
- [3] H. A. Santoso, A. N. Sri Winarsih, E. Mulyanto, G. W. Saraswati, S. E. Sukmana, S. Rustad, M. S. Rohman, A. Nugraha and F. Firdausillah, "Dinus Intelligent Assistance (DINA) Chatbot for University Admission Services," *iSemantic*, 2018.
- [4] D. Hazarika and S. Poria, "Conversational Transfer Learning for Emotion," *arXiv*, 2020.
- [5] Nguyen and M. Luan, "Context-Aware Text Representation for Social Relation," *ACM*, 2016.
- [6] E. Cambria, "Deep Learning-based Document-based Document Modeling for Personality Detection from Text," *IEEE*, 2017.
- [7] B. Kratzwald, S. Ilic, M. Kraus, S. Feuerriegel, and H. Prendinger, "Deep Learning For Affective Computing: Text-based Emotion Recognition in Decision Support," *ArXiv*, 2018.
- [8] P. Liu, . X. Qiu and X. Huang, "Recurrent Neural Network for Text," *IJCAI*, 2016.

- [9] J. Chung and C. Gulcehre, "Empirical Evaluation of," *arXiv*, 2014.
- [10] I. V. Serban, A. Sordoni, Y. Bengio, A. Courville and J. Pineau, "Building End-To-End Dialogue Systems Using Generative Hierarchical Neural Network Models," *arXiv*, 2016.
- [11] N. Radziwill and M. Benton, "Evaluating Quality of Chatbots," *arXiv*, 2017.
- [12] Vishal A. Kharde and S.S. Sonawane, "Sentiment Analysis of Twitter Data: A Survey of Techniques" *arXiv*, 2016.
- [13] Jian Zheng, Cencen Xu, Ziang Zhang and Xiaohua Li, "Electric Load Forecasting in Smart Grid Using Long-Short-Term-Memory based Recurrent Neural Network" researchgate, 2017